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Now-casting and Temporal Disaggregation Dynamic Factor model for Brazilian Quarterly Real GDP

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ABSTRACT

The aim of this work is, firstly, to implement a method of disaggregation of the quarterly Brazilian GDP, estimating a monthly series at constant prices for the period 2003-2020, secondly to obtain predictions of a step forward with superior predictive capacity than the forecasts obtained by the indicator coincident IBC-Br. In order to achieve this goal, different interpolation techniques are tested, all represented in state-space models, and monthly coincident indicators are used as a way to reach the best estimation monthly GDP possible. Based on a robust selection criteria, several models are estimated and the best method of interpolating quarterly GDP is dynamic model with AR (1) errors, estimated by Kalman filter.

Keywords: Kalman Filter, National Accounting, Data Disaggregation, Now-Casting, Dynamic Factor Models, Predictive Accuracy.

■ INTRODUCTION

A constant and systematic monitoring of economic activity is something essential to any country of the world, and the Gross Domestic Product[1] is a measurement that helps economic agents to take important policy decisions in any government.

The aim of this paper is to collaborate with this line of research and to have a monthly measure a measure of economic activity in Brazil that is timely and credible. Economic variables can be found in various frequencies, from financial data available in real time to other important variables that become available on a quarterly, annual, or even decadal, as in the census. These series with lower frequency are available in this way for a variety of reasons: costs associated with collecting data, practical limitations for faster data collection.

This is not different in Brazil. Nevertheless, how to measure economic activity in a monthly frequency and in a timely manner? In the case of Brazil, the country´s GDP is released in a quarterly basis, with a lag of two months from the last month related to the quarter measured, and four months compared to the first month of the quarter analyzed.

Nevertheless, we need the availability of monthly indicator and in a timely manner. The Central Bank of Brazil Economic Activity Index (IBC-Br), first released in 2010, with the beginning of the series starting in 2003, was fundamental to advance monitoring of economic activity in real time. However, is there any room for advancement and improvement? Especially in relation to their predictive performance?

Mainly in academic research, it is common to use the Monthly Industrial Production (PIM-PF), but recently, there has been a gap between the two series: PIM-PF and GDP. Nowcasting is more related to "forecasting the past". In this case, we use monthly data in a timely manner. By doing this, we are able to work "one month ahead" (with a more restricted group of variables) and two months ahead (incorporating IBGE data and IBC-Br).

Our challenge is to come up with a monthly activity indicator that: i) its quarterly value equals the quarterly GDP, i.e., there is the restriction that monthly values calculated sum (or average); ii) the model using "$t + 1$" monthly variables, now-casting produces satisfactory results; ii) the model using "$t+2$" monthly variables, now-casting produces better results than IBC-Br.

In Brazil, which has been constantly attempting to improve its economic activity measures, mainly in relation to filling the gap between the availability of quarterly national accounts released by IBGE, the country´s Bureau of Statistics, with monthly estimates of the national

1 We are fully aware of the problems related to the measurement of GDP, as it does not include important topics, such as environmental ones.

production. Certainly, official monthly data, such as industrial production and retail sales, are able to give us a good idea of what is going on with the country's economic activity. However, an estimate of monthly GDP is needed as any sectorial indicator fails to encompass the gross domestic production as a whole. Moreover, the national accounts data are always available with a considerable lag.

In fact, any type of statistical indicator is of utmost importance in order to design correct economic policies necessary for economic and social development and growth. They are also very important in order to assess whether economic decision have been correctly, or if they need some type of rebuilding.

Through good GDP measurements, policymakers, for instance, are able to make decisions that smooth business cycles and bring the lowest sacrifice ratio possible. Thus, it turns out that the GDP is released on a quarterly basis, with some lag. This is quite plausible since the Brazilian System of National Accounts is complex and requires some time until it is properly measured and, then, released to the public in general.

It happens that an agent´s decision-making process is, in general, faster than the releases of the quarterly GDP.

This makes it necessary to use "proxies" with a frequency greater than serve as a good parameter estimation for the economic activity of the country, or breakdown of robust statistical methods that are able to tell a story of the country's economic activity before the GDP is eventually released.

The aim of this paper is to contribute to literature of temporal disaggregating and now-casting the Brazilian real GDP. This work combines these two lines of research with relevant results for the Brazilian case. This work can be divided into three: i) disaggregation the quarterly Brazilian GDP, estimating a monthly series at constant prices for the period 2003 -2020, ii) Forecast the Brazilian GDP one step ahead using real-time data vintages; iii) measure and test the accuracy of the model forecasts out of sample. The results of the study show that: i) the breakdowns with dynamic methods have superior performance to more traditional methods, ii) Models using Dynamic Factor Temporal Disaggregation have better performance. iii) The predictive ability of the model with may increase using monthly variables "$t + 2$", model with only "$t + 1$" variable or without IBC-Br as co-variable, has the same predictive ability of IBC-Br, iv) all the estimated models have as a byproduct the monthly estimate of real GDP in the same measure of scale of their measurement, with the monthly average of three months of estimated real GDP coinciding with the value of real GDP.

Besides this introduction, this paper is structured as follows. Section 2 situates the theoretical and empirical debate on the topic. Section 3 discusses the methodology to be used at work. Section 4 Procedure for the selection of variables and models. Section 5 reports the data and makes the descriptive analysis of the series used in the study. Section 6 reports the estimation results and the Section 7 presents a nowcasting exercise. Finally, section 8 summarizes findings related to work.

■ LITERATURE

In most countries GDP has a quarterly frequency, these data are often subject to revision, for a paired analysis of forecasts, it is necessary to use the data that was worked at the time of the forecast, which means the use of vintage data. Any forecast nowcasting will necessarily use this type of data. Forecasts for nowcasting a country's output concentrate on real GDP, ie nominal GDP deflated to eliminate price fluctuations. Real GDP in this way becomes an index and can then be quoted in the value of the unit of account for a given year, but in practice the underlying real growth data is directly from the index.

The mixed frequency models presented higher predictive results than the models with only quarterly data. The use of these models leads a research line on it nowcating, which means to predict using all the available data, of an already realized variable, as in the case of GDP that has a lag in the disclosure of results for the current period. The research using these models with mixed frequencies and factorial dynamics has results for GDP forecast for both the American and European cases, as we can relate the studies Marcellino & Schumacher (2010), Camacho & Quiros (2011), Camacho *et al.*(2012), Babura & Ru¨nstler (2011), Giannone *et al.* (2008), Angelini *et al.*(2011), Giannone *et al.* (2004), Trehan (1989), Mariano & Murasawa (2003), Evans (2005) and Proietti & Moauro(2006).

Clements & Galv˜ao (2008) has results where the mixed-frequency AR model performs better than the traditional AR model. In the midline model, the MIDAS model has better predictive capacity in the short term (up to 5 months) than other linear models and the MF-VAR model has a better predictive capacity in the long run (over 9 months). Aruoba *et al.* (2009) and Aruoba & Diebold (2010) use high frequency data in dynamic factorial models.

The studies that analyze several predictions of nowcasting for different countries and situations we highlight Kuzin *et al.*(2013) paper examines the forecast stability of MIDAS and factors models for nowcasting quarterly GDP for six large industrialized countries and find that the nowcast performance of single models varies considerably over time. Hindrayanto *et al.* (2016) examines the performances of four different factor models in a pseudo real-time forecasting competition for the euro area and five of its largest countries concluding that factor models consistently outperform the benchmark autoregressive model. Liu *et al.*(2012) evaluates nowcasts and forecasts of real GDP growth using five models for ten Latin American countries. The results indicated the flow of monthly data helps to improve forecast accuracy, and the dynamic factor model consistently produces more accurate nowcasts and forecasts relative to other model specifications.

Table 1. Country review of now-casting literature.

Country Canada	Model DFM	Description U.S. co-variables	Authors & Year Bragoli & Modugno
U.S.	DFM	Endogenous Structural breaks in mean and variance	Chauvet et al.(201
Mexico	DFM	Novel real-time database	Delajara (2016)
Brazil	TDM-SS	Using 4 monthly co-variables	Notini et al. (201
Germany	Bridge Models	Using seven monthly co-variables	Antipa et al. (201
France	ADL-Midas Blocking Approach	Combination of forecast	Bec & Mogliani (20
Spain Russian Czech Slovakia	DFM DFM DFM	Using 10 monthly co-variables Blocking monthly co-varia-bles Using 27 monthly co-variables	Camacho & Quiros Por-shakovet al. (20 Rusnak (2016) Peter (2017)
Netherlands 0.99	Static, levels, AR(1) 0.89	IP, Rsal	*Chow & Lin* (197
Portugal 0.99	Static, levels, AR(1) 0.89	IP	*Chow & Lin* (197
Spain 0.99	Static, levels, AR(1) 0.89	IP	*Chow & Lin* (197
Brazil -	Static, levels, AR(1) 0.80	IP, Sales	*Chow & Lin* (197

Data interpolation methods can be divided into two groups. There are those that make use of time series models to transform low frequency data into compatible higher frequency ones. This is the case of Boot, Feibes & Lisman (1967). The authors proposed a technique to disaggregate annual data in which, by least squares, they have their first or second differences minimized, subject to annual constraints (Denton, 1971)[2].

On the other hand, a vast interpolation has worked with regression models. One of the main articles in this matter is Chow & Lin (1971), who proposed a direct application of the theory about linear regression model and what is related to a Best Linear Unbiased Estimation. The authors suggested a regression without the addition of dependent lagged variables but with first-order autocorrelated errors.

Since then, several other authors have contributed to the literature related to the disaggregation of time series. Fernandez (1981), for instance, proposed a model estimation similar to Chow Lin´s method, without, however, working with variables in level, but in first difference and, thus, taking into account the possibility of non-stationary processes. Gregoir (1995) and Salazar *et al.* (1997, 1998) took a step forward and ran dynamic regression models as a basis for making the temporal disaggregation of a variable of interest and, therefore,

2 The reader can refer to Pavia-Miralles (2010) for further details on this topic. Proietti (2006), Proietti & Moauro (2006, 2008) are also good references for different methodologies related to temporal disaggregation.

derived estimates based on a framework of a constrained optimization of a quadratic loss function (Fonzo, 2003).

Santos Silva Cardoso (2001) presented a simple application of Chow & Lin´s interpolation method. The idea of the authors was to work with a time series disaggregation method using dynamic models, adding considerable flexibility to the system, particularly when the series used are either stationary or cointegrated. Mitchell *et al.* (2005) proposed formulations related to dynamic regression models. The authors used lagged observations of the variable to be interpolated in the regression equation and applied the model to the British case of the UK.

A further step was the application of traditional models, such as Chow & Lin (1971), in estimations via state space models. Al-Osh (1989) proposed a dynamic linear model, using Kalman Filter estimators, as an approach to the disaggregation of time series. Cush & Hess (2000) used the model proposed by by Al-Osh (1989) for the Swiss GDP, for the quarterly series ranging from 1980 to 1997. Frale *alli*. (2008) use a similar method for the Euro Area GDP and Tasdemir (2008) breaks down the Turkish quarterly GDP for the period between 1989 and 2007. Mönch & Uhlig (2005) use a flexible interpolation routine to disaggregate the quarterly GDP data for the U.S. (1967-2002) and Europe (1970-2005).

For the specific case of Brazil, Cardoso (1981) was one of the first attempts to make a disaggregation the Brazilian GDP. Contador Santos Filho (1987), Nakane (1994) and Cerqueira (2008) also contributed to the literature in some ways.

Notini & Issler (2008), Notini *et alli*. (2012) and Notini & Issler (2016) are more recent attempts to disaggregate the Brazilian GDP, via State Space models. Notini & Issler (2016) argues that the methods of disaggregation using the Kalman filter as estimation methodology has several advantages over other more traditional methods. First, the Kalman filter ensures that the sum of the three months of the series is equal to the unobserved quarterly data. Secondly, the flexibility that the filter allows multiple models to be tested, allowing a better handling of cases of non-stationarity and the different hypotheses waste (Issler & Notini, 2016).

$$y_t = \beta_0 + \beta_1 D + e_t,$$

$$y_t = \beta_0 + \beta_1 D_1 + ... + \beta_{11} D_{11} + e_t,$$

$$t = 1, ..., T.$$

$$y_t = \beta_0 + \beta_1 y_{t-1} + e_t,$$

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_7 y_{t-7} + e_t,$$

$$t = 1, ..., T$$

TEMPORAL DISAGGREGATION METHODS

Following Chow & Lin´s (1971) seminal papers, a variety of other papers have been published in order to come up with solutions for some problems related to Chow Lin procedure. But the best thing a researcher can to is to resort to different methods in order to obtain satisfactory monthly interpolations from quarterly observations (Abeysinghe & Lee, 1998).

There are parametric methods, such as Chow & Lin (1971), which make use of regressions with AR(1) errors.

Chow & Lin (1971) use Feasible OLS and give a numerical solution to estimate the autoregressive component.

Others, such as Santos Silva & Cardoso (2001) make use of dynamic regression models. The authors use a restricted maximum likelihood. They use dynamic components, which are estimated in the region of stationarity (-1,1).

On the other hand, Monch & Uhlig (2005) work with state space models. It was used Kalman Filter recursive maximum likelihood. Chow & Lin´s (1971) can be defined with an initial problem. Firstly, there is a temporal aggregation, i.e., high frequency data converted into low frequency data. Secondly, there is the opposite, i.e., low frequency data converted into high frequency data.

Chow & Lin´s (1971) can be defined with an initial problem. Firstly, there is a temporal aggregation, i.e., high frequency data converted into low frequency data. Secondly, there is the opposite, i.e., low frequency data converted into high frequency data.

We define $y_t = y_{h,t}$: high frequency data in period t (The same sense, $y_{l,t}$: low frequency data in period t).

$$y_{l,t} = \sum_{i=1}^{s} c_i y_{h,i}$$

$$Y_l = CY_h$$

$$Y_h = AY_l$$

Where: “C” is a temporal aggregation matrix and “A” is a temporal disaggregation matrix. or instance, a temporal aggregation matrix related to monthly data converted into quarterly data:

$$C_{t\times 3t} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

If there are covariates that cointegrate with values “h” for high frequency and “l” for low frequency, we might have the following models:

$$Y_h = X_h'\beta + u_h \qquad u_h \sim N(0, V_h) \tag{1}$$

$$Y_l = X_l'\beta + u_l \qquad u_l \sim N(0, V_l) \tag{2}$$

Where *Vh* and *Vl* are respectively the covariance matrix of high frequency data and low frequency data. The relationship between high and low frequency data can be represented by the following disaggregation matrix:

$$\hat{Y_h} = A(X_l'\beta + u_l) \Leftrightarrow \hat{Y_h} = AY_l \tag{3}$$

$$u_{h,t} = \rho u_{h,t-1} + \epsilon_t$$

$$\epsilon \sim N(0, \sigma_\epsilon^2)$$

Thus, an unbiased estimate of *Y^h* can be obtained through the estimation of the disaggregation matrix *A*, it being possible by means of a minimization process of $COV(\hat{Y}_h - Y_l)$. This is the central idea of the article Chow & Lin (1971), which estimates the disaggregation matrix using least Feasible OLS.

Santos Silva & Cardoso (2001) take into account a Dynamic ARMAX model such that:

$$\hat{Y_{l,t}} = C(\Phi Y_{h,t-1} + X_{h,t}'\beta + u_{h,t}) \tag{4}$$

$$u_{h,t} = \rho u_{h,t-1} + \epsilon_t$$

$$\epsilon_t \sim N(0, \sigma_\epsilon)$$

The Santos Silva & Cardoso (2001) method uses a restricted maximum likelihood estimation by the region of stationarity (-1,1)

A Disaggregation Model using Kalman Filter[3]

Following Mönch & Uhlig (2005)[4] we can represent different methods of interpolation in a state-space representation. For instance, Mönch & Uhlig (2005) consider the following dynamics regression model:

$$(1-\phi_1 L - ... - \phi_p(L^p))y_{h,t} = x'_{h,t}\beta + u_{h,t} \tag{5}$$

$$u_{h,t} = \rho u_{h,t-1} + \epsilon_t$$

$$\epsilon_t \sim N(0, \sigma_\epsilon)$$

where $y_{h,t}$ is the high frequency observation of the variable to interpolated, (L^p) is a lag polynomial of order[5] p, $x_{h,t}$ are the time t observations of a set of related series; *ut* is the regression residual which is assumed to follow an MA(1) process.

As in Mönch & Uhlig (2005), we assume that the Brazilian quarterly GDP figures are the average of the unobserved three consecutive monthly observations. Therefore, if we define the quarterly indicator variable y^+:

$$y^+ = (0 \quad 0 \quad y^3 \quad 0 \quad 0 \quad y^6 \quad 0 \quad 0 \quad y^9 ...)$$

As a result, we are able to obtain the following measurement equation:

$$y^+_{l,t} = \begin{cases} \frac{1}{3}\sum_{i=0}^{2} y_{h,t-i}, & t=3,6,9,... \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

There is no error term in the equation above as the mean of three consecutive months is exactly equal to the quarterly observation. We can observe that measurement equation as a matrix of temporal aggregation.

The state-space representation can be as follows:

3 This section is a summary of Appendix A in Mönch & Uhlig (2005).

4 Their procedure has been used by Issler & Notini (2008) and Notini et alli. (2012) for the Brazilian data.

5 We follow Mönch & Uhlig (2005) and, for simplicity, we assume that p = 1. Otherwise, we would generate a great number of different models to be compared.

$$y_{l,t}^{+} = H_t \xi_t \tag{7}$$

Where:

$$H_t = \begin{cases} [1/3\ 1/3\ 1/3\ 0], & \text{para, } t = 3, 6, 9, \ldots \\ [0\ 0\ 0\ 0], & \text{otherwise} \end{cases}$$

$$H_t = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xi_t = \begin{bmatrix} y_{h,t} \\ y_{h,t-1} \\ y_{h,t-2} \\ u_{h.t} \end{bmatrix} = \begin{bmatrix} \phi & 0 & 0 & \rho \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \rho \end{bmatrix} \times \begin{bmatrix} y_{h,t-1} \\ y_{h,t-2} \\ y_{h,t-3} \\ u_{h,t-1} \end{bmatrix} + \begin{bmatrix} X'_{h,t}\beta \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \epsilon_t \\ 0 \\ 0 \\ \epsilon_t \end{bmatrix}$$

Mönch & Uhlig (2005) show that the model reported above is capable of dealing with various types of interpolation models, by fixing either ϕ or ρ. For instance, the seminal Chow & Lin (1971) methodology, which is a regression model with no lagged dependent variables, but with autoregressive errors, can be obtained by fixing $\phi = 0$ and letting the parameter ρ free. In order to take into account cases of nonstationarity in the data, Fernandez (1981) works with a model in first differences. This can be obtained by making $\phi = 0$ and $\rho = 1$, i.e., the regression residuals will be treated as a random walk process. Mitchell *et al.* (2005), on its turn, work with a dynamic regression model with IID errors. This can be obtained by letting $\phi \neq 0$ e $\rho = 0$. Table 1 reports the parameter restrictions imposed by various interpolation methods.

Table 2. Models and Specifications Using State-Space Structure - Source: Mönch & Uhlig (2005) and Authors.

Nº	Model	Φ	ρ
M1	Static Model in levels with IID residuals	0	0
M2	Static Model in levels with AR(1) residuals (*Chow & Lin (1971)*)	0	free
M3	Static Model in first differences with IID residuals (*Fernandez (1981)*)	0	1
M4	Dynamic Model in levels with IID residuals (*Mitchell et al (2005)*)	free	0
M5	Dynamic Model in first differences with IID residuals	free	1
M6	Dynamic Model in levels with AR(1) residuals (*Santos Silva e Cardoso (2001)*)	free	free

In the context of estimation, the idea is to estimate via Kalman filter monthly gdp through recursive maxi- mum likelihood with the constraint equation (6). The "nowcasting" predictions will be performed using *smoothed* estimate: using only observations available up to period "t – 1", thus we have the following temporal sample (y_1, x_{1},$_{1}$, y_2, x_{1},$_{2}$, ..., yt_{-1}, xk,t_{-1}).

MODEL SELECTION CRITERIA: PROCEDURE FOR THE SELECTION OF VARIABLES AND MODELS

Initially, our set of variables had 54 variables in total. The process of selecting variables considered the following steps:

1. Unit root tests: ADF, MADFgls, with and without structural breaks Perron (1989), determinist trend with structural breaks;
2. Cross correlation in first difference as been applied by Monch Uhlig (2005), Notini Issler (2008) and Notini et alli. (2012);
3. In addition to Granger causality tests and cross-correlations, some autoregressive systems were examined to assess the marginal predictive content of the variables for GDP;
4. Cross-correlations and Principal Component Analysis[6], was considered for final selection;

The procedure undertaken here is similar to the NBER approach and the one pursued by Stock and Watson (1989, 1991) and Chauvet (2016, 2002, 2001), which list a large number of variables and reach a shorter list of variables that enter their leading indicators.

After that, we estimated a general model (via Kalman filter) with different combinations of variables selected previously. The criteria involved three methods: MAPE, RMSE and R^2.

In order to evaluate the quality of the interpolation processes, we apply three methods of evaluation. The first one follows Bernanke, Gertler & Watson (1997) and uses R2 measures of fit[7].

6 consider this treatment for the series in first difference.

7 M¨onch & Uhlig (2005) use the same method.

Denote $y^{+}_{t|T}$ the GDP´s monthly expected value related to period "*t*" and conditional on the estimated model parameters and the full information set.

$$R^2_{level} = \frac{Var(y^{+}_{t|T})}{Var(y^{+}_{t|T}) + Var(u_{t|T})}$$

As we know, the presence of variable trends in economic time series may lead to wrong conclusions. In order to avoid this problem with the variables to be interpolated, as well as the variables used in the disaggregation, it is more informative to report the R^2 in first differences to consider the quality of fit within the sample and minimize or even eliminate the effects of those trends.

$$R^2_{diff} = \frac{Var(\Delta y^{+}_{t|T})}{Var(\Delta y^{+}_{t|T}) + Var(\Delta u_{t|T})}$$

The mean absolute percentage error (MAPE) assesses forecast accuracy and it expresses such accuracy in percentage terms, which it makes much easier to understand. This is very important once, as mentioned by Proietti (2006), the choice of a model has to take into consideration its forecast capability.

The MAPE is defined as follows:

$$MAPE = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\hat{y^{+}_t} - y^{+}_t}{y^{+}_t}\right|$$

The Root Mean Square Error (RMSE) measures the differences (the residuals) between the values predicted by a model and those actually observed. Those residuals are aggregated into a single measure of predictive power and, as a result, the RMSE penalizes large prediction errors.

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{y^{+}_t} - y^{+}_t)^2}$$

DATA, DESCRIPTIVE STATISTICS AND TESTS

The first procedure for a better data disaggregation is to find high-frequency variables capable of serving as good indicators for data interpolation. The definition of these coincident variables dates back to Burns and Mitchell (1946) and, basically, they are high-frequency real time variables, highly correlated with the countrys business cycle. These leading indicators should certainly have a considerable co-movement with the quarterly GDP in order to have the best possible interpolation, as well as to come up with a good indicator of the countrys business cycle.

Period under analysis, ranging from the 1st quarter of 2003 to the 4rd quarter of 2020, a smaller data range is offset by a greater availability of variables. Therefore, the estimation of the models proposed in this work will involve monthly data for the period 2003-2020. The variables used in the estimations proposals are listed below[8]:

Table 3. Variables Selected.

xi	Variable	Availability	Unit	Source
1	Capacity Utilization Level (NUCI)	t + 1	index	FGV
2	Present Situation Index (ICI)	t + 1	index	FGV
3	Expectations Index (IE)	t + 1	index	FGV
4	ABCR Heavy Vehicle Traffic on Tollroads	t + 1	thousands	ABCR
5	Average Electric Energy Consumption	t + 1	Mwmed	ONS
6	M1 (deflated by IPCA)	t + 1	millions	BACEN
7	Credit Protection Service (SPC)	t + 1	number of queries	ACSP
8	Agricultural Activity Index	t + 1	index	MF
9	ABPO Corrugated Fiberboard and Cardboard Sales	t + 1	thousand tons	ABPO
10	Cement Production	t + 1	tons	SNIC
11	Inventory levels	t + 1	index	FGV
12	Import Quantum	t + 1	index	FGV
13	Export Quantum	t + 1	index	FGV
14	PMC - Monthly Retail Survey	t + 2	index	IBGE
15	PIM - Monthly Industrial Production	t + 2	index	IBGE
16	IBC-Br - Brazil Central Bank Economic Activity Index	t + 2	index	BACEN

With this selected variables, will be tested a number of models previously described and evaluated their adjustment and their predictive ability. Subsequently selecting a model, an “nowcasting” exercise will be implemented. We emphasize that some variables have a greater lag of publication of results, particularly in the case of this study, these variables are crucial in the estimation and forecasting of monthly GDP: PMC, PIM and IBC-Br. Thus, a model that will be proposed, include only variables with dynamic “t + 1”, and prior to the final estimation will be implemented for each variable, the extent to which the same become available, following respectively: PIM, PMC and IBC-Br. The table (3)[9] we have results for the unit root tests considering the series of high frequency. Consider a comparison we present two tests for this purpose: the classic ADF and *MADFGLS* to treat low power issue.

8 These variables have been selected according to the description in section 4

9 In ADF Test. the specification of the equation was chosen on the basis of the Schwarz Information Criterion; ii) We consider 14 as maximum lag as 14; iii) Drift and linear trend was included in the best equations.

Most parts of the results of unit root tests for high frequency data (monthly), have converging results. However it should be emphasized that some conflicting results may be caused by the presence of structural breaks, which in this study will not be treated. Most series have a unit root but are stationary in first difference[10]. We present in Table (4)[11] the same unit root tests yet to the quarterly series. Considered to analyze the GDP series, three different temporal samples: Since the beginning of the serie (since 1991), from 1996 and in the case of the temporal dimension of this study, from 2003.

The results of unit root tests for quarterly data are similar to monthly returns (for variables with two dynamic). For the most part, the results of the tests are convergent. However, the number of divergences in results between the ADF and MADF tests, increases in the quarterly case, especially by reducing the time dimension of the sample and the possible presence of structural breaks. The GDP also shows divergent results. The study will follow, assuming as valid the results of stationarity of the series, particularly in the period from 2003. We present in table (5)[12] the results of the quarterly correlations for the series in level, first difference and in first difference with seasonal adjustment.

Table 4. Quarterly Correlations.

N◦	Variables	$\rho(GDP,xi)$	$\rho(\Delta GDP,\Delta xi)$
1	Capacity Utilization Level (NUCI)	0.47	0.73
2	Present Situation Index (ICI)	0.37	0.61
3	Expectations Index (IE)	0.29	0.62
4	ABCR Heavy Vehicle Traffic on Tollroads	0.96	0.88
5	Average Electric Energy Consumption	0.93	-0.62
6	M1 (deflated by IPCA)	0.96	-0.28
7	Credit Protection Service (SPC)	0.87	0.47
8	Agricultural Activity Index	0.40	0.10
9	ABPO Corrugated Fiberboard and Cardboard Sales	0.95	0.88
10	Cement Production	0.98	0.74
11	Inventory levels	0.50	0.72
12	Import Quantum	0.98	0.69
13	Export Quantum	0.66	0.87
14	PMC - Monthly Retail Survey	0.94	0.40
15	PIM - Monthly Industrial Production	0.88	0.90
16	IBC-Br - Brazil Central Bank Economic Activity Index	0.99	0.96

Source: authors elaboration.

10 The resutlados for the unit root test for the first were suppressed, and are available upon request.

11 We have to aggregate the monthly series in order to compute the tests. i) In ADF Test. the specification of the equation was chosen on the basis of the Schwarz Information Criterion; ii) We consider 14 as maximum lag as 14; iii) Drift and linear trend was included in the best equations.

12 We also compute the correlation coefficient between each candidate series cycles (extracted by Hodrick-Prescott filter) and GDP cycle. Results are similar to those in table (5)

The results of the quarterly correlation with GDP, indicate that all these variables have considerable correlation, either in level or first difference. A relevant question in this modeling is the identification of the structure of autocorrelation in the series of GDP. This information can indicate whether a dynamic model may be appropriate. Accordingly, presented in table (1), the results of tests for the autocorrelation series and the first level difference.

Table 5. Autocorrelation analysis

Autocorrelation analysis of the GDP series				
	AC	**PAC**	**Q-Stat**	**P-value**
1	0.912	0.912	59.076	0.000
2	0.827	-0.026	108.41	0.000
3	0.747	-0.021	149.23	0.000
4	0.665	-0.049	182.15	0.000
5	0.590	-0.016	208.43	0.000
Autocorrelation analysis of the GDP series - First difference				
	AC	PAC	Q-Stat	P-value
1	0.204	0.204	2.903	0.078
2	-0.003	-0.046	2.903	0.234
3	-0.113	-0.108	3.826	0.281
4	-0.199	-0.162	6.730	0.151
5	-0.034	0.037	6.817	0.235

Included observation:
Sample: 1996Q1-2020Q4

Source: authors elaboration.

The dynamics presented by GDP, indicates the possibility of using dynamic models of up to order 2 will focus, as well as Mönch & Uhlig (2005), our study to the case of the first order.

The approach proposed by Mönch & Uhlig (2005) was used to model the GDP of several nations[13], each nation having its own specification. Generally, using only one or two sets of covariates. The most common specification uses an AR (1) static model, with variable level, representing the Chow & Lin (1971) model in a structure of state-space. As shown in table (6):

Table 6. Interpolation specification, related series used for interpolation, goodness of fit.

Country	Model	Rel. Series	Authors	R^2_{level}	R^2_{diff}
Austria	Dynamic, levels, AR(1)	IP, Empl	*Santos Silva e Cardoso* (2001)	0.99	0.49
Belgium	Static, 1st diffs, IID	IP, Rsal	*Fernandez* (1981)	0.99	0.89
Finland	Static, levels, AR(1)	IP, Rsal	*Chow & Lin* (1971)	0.99	0.89
France	Static, levels, AR(1)	IP	*Chow & Lin* (1971)	0.99	0.89
Germany	Static, levels, AR(1)	IP, Rsal	*Chow & Lin* (1971)	0.99	0.89
Greece	Static, levels, AR(1)	IP, Rsal	*Chow & Lin* (1971)	0.99	0.89
Ireland	Dynamic, levels, IID	Rsal, Expts	*Mitchell et al* (2005)	0.99	0.89
Italy	Static, levels, AR(1)	IP	*Chow & Lin* (1971)	0.99	0.89

13 Most results for European countries used the data up to the year 1995.

Country	Model	Rel. Series	Authors	R^2_{level}	R^2_{diff}
Luxembourg	Dynamic, levels, IID	IP, Empl	*Mitchell et al* (2005)	0.99	0.89
Netherlands	Static, levels, AR(1)	IP, Rsal	*Chow & Lin* (1971)	0.99	0.89
Portugal	Static, levels, AR(1)	IP	*Chow & Lin* (1971)	0.99	0.89
Spain	Static, levels, AR(1)	IP	*Chow & Lin* (1971)	0.99	0.89
Brazil	Static, levels, AR(1)	IP, Sales	*Chow & Lin* (1971)	-	0.80

Source: Mönch & Uhlig (2005) and Authors.

Although the time series considered, have a smaller sample size, has variables such sample with a more informative set for nowcasting weather, especially in recursive estimations where the point of beginning of the estimation is as relevant as the temporal dimension, as we shall see in the next section. Other studies used a smaller set of covariates, with greater temporal dimension, which can impair the predictive ability of the model.

RESULTS OF MODEL SELECTION: CONCLUSION

In this section we present the estimates of models for the various table options (1) besides the SSC model described in section (3). The choice of model had taken into account the combination of the measures listed in section (4). ted below in figure (2) the result of this process.

Model Selection					
Models	MAPE	RMSE	R^2	Description	Variables Used
M(T+2)	0.0015	0.0061	0.7742	M6 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$
M.IBC-Br	0.0025	0.0099	0.7666	M6 - Monch & Uhlig - Table (1)	$t + 2$ without IBC-Br and $t + 1$
M(T+1)	0.0035	0.0171	0.6274	M6 - Monch & Uhlig - Table (1)	$t + 1$
M1	0.0018	0.0362	0.8490	M6 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$. replacing ABPO by income tax
M2	0.0061	0.0119	0.9490	Santos. Silva & Cardoso (2001) - Section (3)	$t + 2$ and $t + 1$
M2.IBC-Br	0.0062	0.0384	0.6490	Santos. Silva & Cardoso (2001) - Section (3)	$t + 2$ without IBC-Br and $t + 1$
M3	0.1377	0.5126	0.2119	M5 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$
M4	0.0687	0.2898	0.6944	M4 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$
M5	0.1456	0.9898	0.2098	M3 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$
M6	0.1311	2.1216	0.7013	M2 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$
M7	0.1111	1.7996	0.6971	M1 - Monch & Uhlig - Table (1)	$t + 2$ and $t + 1$

Source: authors elaboration. Recursively estimated one step a head, in levels, without seasonal adjustment.

The specifications that consider dynamic models (both Kalman filter as Santo Silva and Cardoso (2001)) presented superior performance both in terms of predictive ability (MAPE and RMSE) and in terms of R^2.The model containing only variables available in $T + 1$ has much lower performance compared to models containing information available in $T + 2$ (industrial production, retail survey and IBC-Br). The results indicate considerable predictive gain when we include the number of IBC-Br.

Period	Rate of change ($t/t-1$) seasonally adjusted					Forecast Error (GDP – Nowcast)			
	GDP*	IBC-Br	T + 1**	T + 2**	T + 2**$_F$	IBC-Br	T + 1**	T + 2**	T + 2**$_F$
2018Q3	0,91	1,14	0,63	0,60	0,57	-0,23	0,28	0,31	0,34
2018Q4	-0,36	-0,13	-0,74	-0,76	-0,50	-0,23	0,38	0,40	0,15
2019Q1	0,91	0,12	0,53	0,29	0,92	0,80	0,38	0,63	-0,01
2019Q2	0,39	0,53	0,23	0,31	0,32	-0,13	0,16	0,08	0,07
2019Q3	-0,11	-0,46	-0,53	-0,79	-0,15	0,35	0,42	0,68	0,04
2019Q4	0,37	0,47	0,42	0,65	0,35	-0,11	-0,06	-0,29	0,02
2020Q1	-2,06	-1,38	-2,04	-2,16	-2,00	-0,68	-0,01	0,10	-0,06
2020Q2	-9,22	-9,98	-8,95	-8,40	-9,13	0,77	-0,27	-0,82	-0,08
2020Q3	7,66	8,26	8,20	8,38	8,00	-0,59	-0,54	-0,71	-0,34
2020Q4	3,16	3,39	3,29	2,96	3,09	-0,22	-0,13	0,21	0,07
Mean Absolute Error						0.41	0.26	0.42	0.12
Mean Square Error						0.23	0.09	0.24	0.03

* Using vintage correspondent.
** Using rolling scheme.
$T+2^{**}$: Using all model variables except IBC-Br.
$T+2^{**}_F$: Using all model variables.

Source: authors elaboration.

Dielbod-Mariano test statistics are asymptotically distributed as a standard normal random variable. The Giacomini- White test statistic is asymptotically distributed as a chi-squared random variable with one degree of freedom. The results show the existence of space for a predictive gain nowcasting fo GDP compared to the simple use of IBC-Br, specially with respect of $T+2$ model. The models $T+2$ without IBC-Br and $T+1$ have at least, the predictive performance equivalent to the IBC-Br but an additional advantage of that is closure of the monthly average in the value of GDP, which does not occur with IBC-Br.

Predictive Ability Tests - IBC-Br			
Test*	$T+2$ and IBC-Br	$T+2^{**}_F$ and IBC-Br	$T+1$ and IBC-Br
GW	0.01	0.05	0.08
DM	0.04	0.07	0.10

*P-value of each test with H_0: Equivalent predictive ability and
HA: first model presents superior predictive ability.
**without IBC-Br

Source: authors elaboration.

The results of the predictive ability tests indicate evidence of greater predictive ability of the models with $T+2$ and $T+2^{**}_F$ compared to the $T+1$ model. The inclusion or not of the IBC-Br does not indicate a statistically significant predictive gain. The results reinforce the possibility of using nowcasting even without using the IBC-Br, with predictive results comparable, or even better to those presented by Notini & Issler (2016).

Predictive Ability Tests - Models			
Test*	T+2 and T+1	$T+2^{**}_F$ and T+1	T+2 and $T+2^{**}_F$
GW	0.01	0.04	0.08
DM	0.05	0.06	0.16

*P-value of each test with H_0: Equivalent predictive ability and

HA: first model presents superior predictive ability.
**without IBC-Br

Source: authors elaboration.

NOWCASTING EXERCISES: "PREDICTING" THE PREVIOUS QUARTER GDP

According to Banbura, Giannone & Reichlin (2010), nowcasting is the prediction of the present, the very near future and the very recent past. Nowcasting is much more than a *"simple production of an early estimate as it essentially requires the assessment of the impact of new data on the subsequent forecast revisions for the target variable"*. This is what we are trying to do with Brazil´s GDP. This is the "prediction" of GDP with information of the variables used in the model for the three months of the previous quarter. For instance, in order to predict the GDP related to the 2th quarter 2020, we use the data related to the selected variables for the months of October, November and December, in addition to the GDP related to the 1rd quarter 2020. Three models used:

1. Model $t + 1$: uses only the variables available in month $t + 1$;
2. Model $t + 2$ without IBC-Br: uses all the variables in the model, except the IBC-Br;
3. Model $t + 2_F$: all model variables.

At this stage, the GDP for the previous quarter was known, as well as the covariates for the current quarter. To elucidate this aspect, we present the graphs of the analysis out-of-sample[14] from which are generated the measures MAPE and RMSE:

14 The analysis out-of-sample prediction was performed with the ten periods, always analyzing the prediction of one step forward, beginning in Q2 2018 to Q4 2020.

Figure 1. Out-of-Sample Analysis: Nowcasting Brazil´s GDP (in levels, without seasonal adjustment) Model $t + 1$.

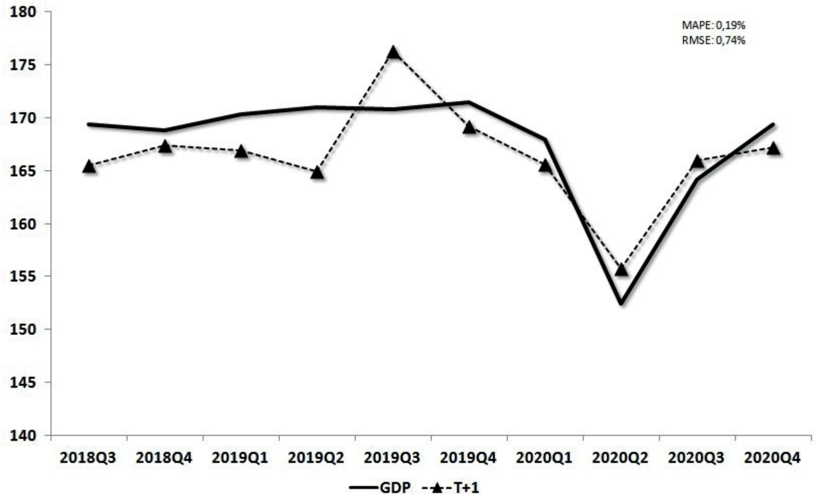


Figure 2. Out-of-Sample Analysis: Nowcasting Brazil´s GDP (in levels, without seasonal adjustment) Model $t + 2$ without IBC-Br.

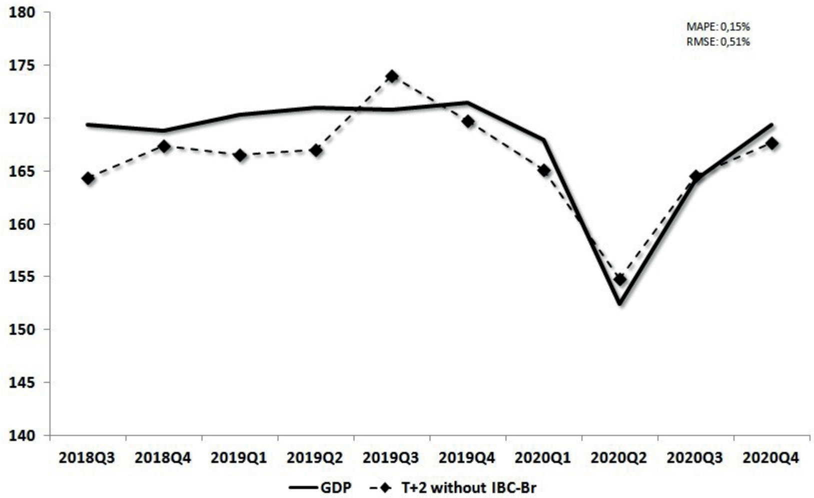
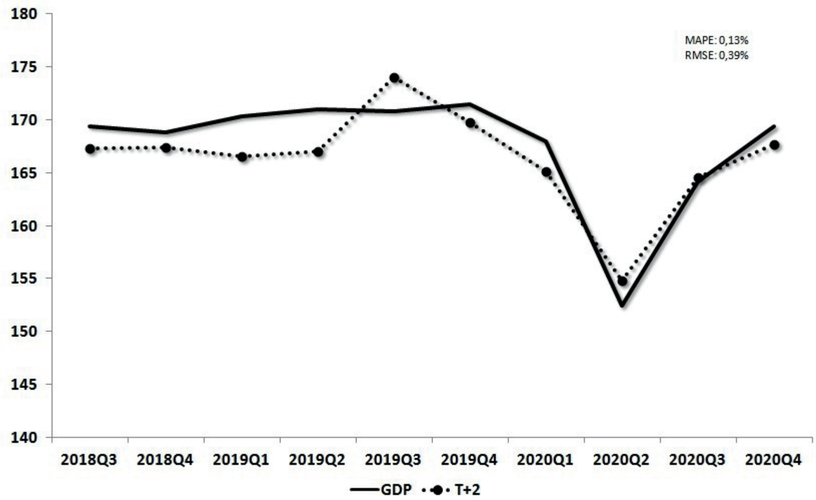


Figure 3. Out-of-Sample Analysis: Nowcasting Brazil´s GDP (in levels, without seasonal adjustment) Model $t + 2$.



Model $t + 2$ predicts better, but it is only available after IBC-Br series is released (around the 16th of the month $t + 2$). In order to overcome this problem, we can calculate some “flash” estimates of important activity indicators, making use of all available data for each moment of time:

1st “flash” estimate: calculated in $t + 1$, probably soon after Funcex (trade balance) is released

2nd “flash” estimate: calculated in $t + 2$, when the PIM (industrial production) is released

3rd “flash” estimate: calculated in $t + 2$, when the PMC (retail survey) is released

Final estimate: calculated in $t + 2$, when the IBC-Br is released

CONCLUSION

The aim of this article was to implement a method of disaggregation of the quarterly Brazilian GDP, estimating a monthly series at constant prices for the period 2003-2020. For this, different techniques are tested interpolation, all represented in state-space models, and monthly coincident indicators are used as a way to get to a better estimation of monthly GDP. Based on a robust selection criteria optimal, several models were estimated and the best one selected was dynamic model with AR(1) errors, estimated by Kalman filter. The estimations performed present results much better than the monthly activity indicator released by the Central Bank of Brazil (IBC-Br), with the advantage that model estimates have the quarterly average coinciding with the real GDP value, that is, on the same measurement scale, which is not possible in the case of IBC-Br. Besides that, the “flash” estimates used enable us to make use of a broader range of variables. But another advantage of the models used in this article is the possibility to included new variables as they become available.

A promising line of development will be the use of data from the Big Data context as used for example by Koop and Onorante (2013), which can probably generate the condition of having more variables than temporal observations, which leads us to use the high-dimensional time series models, already discussed by Belloni and Chernozhukov (2009).

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